Program: **Diploma in Civil & Environmental Engineering**	
Course Code: **5351**	Course Title: **Environmental Engineering**
Semester: **5**	Credits: **4**
Course Category: **Program Core Course**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- To understand of environmental engineering principles and their application in civil engineering projects.
- To explain the water and wastewater management, solid waste management, environmental pollution, and sustainable engineering practices.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Fluid Mechanics	3352	Mechanics of Fluids	3
Basic knowledge of Building Materials	3357	Building Materials and Construction Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the principles and processes involved in water treatment	15	Understanding
CO2	Describe various treatment processes used in wastewater treatment plants.	15	Applying
CO3	Describe the design of water treatment and waste-water treatment plants.	14	Understanding
CO4	Explain various methods of solid waste management and air pollution control.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

<table>
<tr><th>Module Outcomes</th><th>Description</th><th>Duration (Hours)</th><th>Cognitive Level</th></tr>
<tr><td>CO1</td><td colspan="3">Explain the principles and processes involved in water treatment</td></tr>
<tr><td>M1.01</td><td>Describe the various physical, chemical, and biological parameters used in assessing water quality.</td><td>7</td><td>Understanding</td></tr>
<tr><td>M1.02</td><td>Understand processes involved in water treatment.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M1.03</td><td>Explain the sources of water for consumption and water demand.</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Sources of water: Surface water and groundwater – types, advantages and disadvantages.
Water demand – definition of different terms, methods of population forecast (only theory and no numerical problems required), relevance of logistic curve.
Water quality and standards: Physical, chemical, and biological characteristics – methods and tests (test conditions, expected outcomes and permissible limits only; detailed procedures not required)
Water treatment processes: Screening, coagulation, flocculation, sedimentation, filtration, and disinfection. (theory and methods only; no design required)</td></tr>
<tr><td>CO2</td><td colspan="3">Describe various treatment processes used in wastewater treatment plants.</td></tr>
<tr><td>M2.01</td><td>Apply the theory of BOD to study the waste water characteristics.</td><td>7</td><td>Applying</td></tr>
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<tr><td>M2.02</td><td>Explain the characteristics of wastewater and their impact on the environment.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Explain the sources and types of waste water.</td><td>3</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test -I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Sources and types of wastewater: Domestic, industrial, and stormwater – definitions and characteristics.
Characteristics of wastewater: BOD, COD and TSS (Definitions and theory). Simple numerical problems related to BOD, dilution factor and deoxygenation constant
Wastewater treatment processes: Preliminary treatment, primary treatment, secondary treatment (aeration tanks, activated sludge process), and tertiary treatment (theory and methods only; no design required).</td></tr>
<tr><td>CO3</td><td colspan="3">Describe the design of water treatment and waste-water treatment plants.</td></tr>
<tr><td>M3.01</td><td>Identify basic water treatment systems, including components.</td><td>6</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Explain the Distribution systems in water supply.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Describe the design of wastewater treatment units.</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Design of water treatment plants: Flow diagrams, components, and capacities (theory only including methods and processes)
Distribution systems: Pipe networks, pumps, and valves (methods and types only)
Case study: Real-life water treatment plant examples (not for direct assessment).
Design of wastewater treatment plants: Flow diagrams (theory only including methods and processes).
Case study: Real-life wastewater treatment plant design (not for direct assessment).</td></tr>
<tr><td>CO4</td><td colspan="3">Explain various methods of solid waste management and air pollution control</td></tr>
<tr><td>M4.01</td><td>Identify Solid waste collection systemsand types of air pollutants</td><td>4</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>Evaluate various solid waste disposal methods.</td><td>6</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Explain the effects of air pollution on human health and the environment.</td><td>4</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – II</td><td>1</td><td></td></tr>
</table>

Contents: Types of solid waste: Municipal, industrial, hazardous, and biomedical waste (definitions only). Solid waste collection systems: Methods of collection, transportation, and handling (theory only). Solid waste disposal methods: Sanitary landfills, incineration, composting, recycling, and waste-to-energy technologies (theory and methods only; no design required). Environmental impact of waste disposal: Groundwater contamination, air pollution, and public health concerns (theory only). Case study: Waste management practices in urban areas (not for direct assessment). Sources of air pollution: Industrial emissions, vehicular pollution, and natural sources (definition only) Effects of air pollution: Health effects, global warming, and ozone depletion(theory only). Ambient air quality monitoring: Techniques and devices (definitions only). Case study: Air pollution control in industrial areas(not for direct assessment).

Text / Reference:

T/R	Book Title/Author
T1	Environmental Engineering: Water, Wastewater, and Solid Waste Management by S. K. Garg
T2	Introduction to Environmental Engineering and Science by Gilbert M. Masters
R1	Wastewater Engineering: Treatment and Resource Recovery by Metcalf & Eddy
R2	Environmental Engineering: A Design Approach by K. N. Duggal

Online resources:

Sl.No.	Website Link
1	https://www.nptel.ac.in/